

A Counterexample to TLMC Conjecture 00000001136: All Bott–Samelson Varieties Are Gorenstein

TLMC falsification package

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Abstract

We disprove TLMC conjecture 00000001136, which asserts that a Bott–Samelson variety $BS(s_1, \dots, s_k)$ is Gorenstein *if and only if* every letter of the word $s_1 \cdots s_k$ occurs equally often. The longest element $w_0 \in S_3$ provides a reduced word $s_1 s_2 s_1$ with letter counts $(2, 1)$; nevertheless $BS(s_1, s_2, s_1)$ is Gorenstein, because every Bott–Samelson variety is an iterated $\mathbb{P}^1$ -bundle, hence smooth projective, and smoothness implies the Gorenstein property. The “only if” direction fails.

1 The conjecture

We quote the original text verbatim.

Conjecture 00000001136. Definition: *Complete intersection property of Bott–Samelson varieties means some power of the canonical bundle is very ample and the cohomology ring is completely determined by Chern classes.* Conjecture: *$BS(s_1, \dots, s_k)$ is Gorenstein if and only if each letter of the word $s_1 \cdots s_k$ occurs equally often; and in the Gorenstein case, the shift of the dualizing sheaf is given by reversal of the word.*

2 The geometric input: smoothness implies Gorenstein

Observation 1. *For a reduced word $s_1, \dots, s_k$, the Bott–Samelson variety $BS(s_1, \dots, s_k)$ is an iterated $\mathbb{P}^1$ -bundle (Demazure, Désingularisation des variétés de Schubert généralisées, *Ann. Sci. ENS* 7 (1974)). In particular it is smooth and projective.*

Observation 2. *Every smooth variety is Gorenstein: smoothness means all local rings are regular local rings, and regular local rings are Gorenstein (standard commutative algebra; e.g. Matsumura, Commutative Ring Theory, §19). Equivalently, the dualizing sheaf of a smooth variety is a line bundle (the canonical bundle).*

Consequently *every* Bott–Samelson variety $BS(s_1, \dots, s_k)$ built from a reduced word is Gorenstein, independently of how often each letter occurs in the word.

3 The counterexample

Let $w = s_1 s_2 s_1$, a word in the simple reflections of the symmetric group S_3 (letters applied left to right). Direct computation:

$$(s_1 s_2 s_1)(1) = 3, \quad (s_1 s_2 s_1)(2) = 2, \quad (s_1 s_2 s_1)(3) = 1,$$

so $s_1 s_2 s_1 = w_0 = (3, 2, 1)$, the longest element of S_3 .

- **The word is reduced.** Its length is 3, and the Coxeter length of w_0 (the number of inversions of the one-line notation $(3, 2, 1)$) is $\binom{3}{2} = 3$. Since $s_1 s_2 s_1 = w_0$ and the lengths agree, the word is a reduced expression.
- **The letter counts are unequal.** s_1 occurs 2 times, s_2 occurs 1 time: $(2, 1)$ with $2 \neq 1$.
- **Nevertheless $BS(s_1, s_2, s_1)$ is Gorenstein,** by Observations 1 and 2: it is an iterated $\mathbb{P}^1$ -bundle, hence smooth projective, hence Gorenstein.

Thus $BS(s_1, s_2, s_1)$ is Gorenstein although the letters of $s_1 s_2 s_1$ do not occur equally often: the “only if” direction of Conjecture 00000001136 fails, and the conjecture is **disproved**.

Observation 3 (Robustness). *The longest element $w_0 \in S_3$ has exactly two reduced expressions, $s_1 s_2 s_1$ and $s_2 s_1 s_2$, with letter counts $(2, 1)$ and $(1, 2)$; no rewording of w_0 satisfies the conjecture’s hypothesis. Worse, since every Bott–Samelson variety is Gorenstein, the conjecture’s “only if” direction would force every reduced word to have equally often occurring letters, which fails already for s_1 (counts $(1, 0)$) in S_3 . With the “iff” false, the claimed “reversal of the word” description of the dualizing shift is moot (for a smooth variety the dualizing sheaf is simply the canonical bundle).*

4 Verification

- `reproduce.py` recomputes all numbers: the evaluation $s_1 s_2 s_1 = w_0$, reducedness (word length 3 = Coxeter length 3), the letter counts $(2, 1)$, and an exhaustive enumeration of the reduced expressions of w_0 in S_3 .
- `lean4/Main.lean` formalizes the disproof in bare Lean 4 core (no Mathlib): letters are `Nat`, the reflections s_1, s_2 and $w_0 \in S_3$ are concrete functions $\mathbb{N} \rightarrow \mathbb{N}$, the word $[1, 2, 1]$ is verified to evaluate to w_0 and to be reduced, and the main theorem `conjecture_1136_false` shows that for *any* predicate G of Gorenstein-ness satisfying the classical hypothesis $\forall w, G w$ (every Bott–Samelson variety is Gorenstein), the conjecture’s biconditional $\forall w, G w \leftrightarrow \text{EqualCounts } w$ is false, witnessed by $[1, 2, 1]$.
- `lean4/Check.lean` audits all six theorems with `#print axioms`: every one depends on no axioms (no `sorry`, no `Classical.choice`, no `propext`, no `Quot.sound`).